

IAS PRELIMS/MAINS



SCIENCE & TECHNOLOGY



A COMPREHENSIVE COVERAGE

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NUCLEAR TECHNOLOGY

INDIAN NUCLEAR DIPLOMACY

HISTORY OF NUCLEAR COOPERATION

- The US helped in the construction of two Boiling Water Reactors (BWR) of 210 MW each at Tarapur Atomic Power Station, Maharashtra. **But the US and Canada terminated their assistance after the detonation of India's first nuclear bomb in 1974 (Operation Smiling Buddha).**
- **The Treaty on the Non-Proliferation of Nuclear Weapons (NPT)** was established to prevent the spread of nuclear weapons and weapon technology, to promote cooperation in the peaceful uses of nuclear energy and to further the goal of achieving nuclear disarmament. **India refused to sign the treaty on the grounds of discriminatory and unfair treatment.**



INDIAN NUCLEAR DIPLOMACY

NUCLEAR SUPPLIERS GROUP (NSG)

- NSG is a **group of nuclear supplier countries** that seek to prevent nuclear proliferation by controlling the export of materials, equipment, and technology that can be used to manufacture nuclear weapons.
- NSG was formed with the objective of **averting the proliferation of nuclear weapons** and preventing acts of nuclear terrorism.
- Members of the NSG - **NSG consists of 48 members** which include the **five nuclear weapon states US, UK, France, China, and Russia**. It is not a formal organization, and its guidelines are not binding. **Decisions**, including on membership, are **made by consensus**.
- In September 2008, the NSG participating governments agreed to grant India a "clean waiver" from its existing rules, which forbid nuclear trade with a country which has not signed the Nuclear Non-Proliferation Treaty (NPT).

Nuclear Suppliers Group



INDIAN NUCLEAR DIPLOMACY

NUCLEAR AGREEMENT WITH THE USA

- India and USA signed Agreement 123 for civil nuclear cooperation.
- After NSG's waiver to India to commence civilian nuclear trade in 2008, the US signed agreement with India on civil nuclear cooperation and even convinced its allies to do so. **Now India is the only country outside the NSG's members who possesses nuclear weapons but is allowed to trade in nuclear materials.**
- The agreement ended technology denial regimes against India that have been in place for three decades and ended India's nuclear isolation.



INDIAN NUCLEAR DIPLOMACY

NUCLEAR AGREEMENT WITH THE US

- The agreement opened the doors for India to have civil nuclear cooperation as an equal partner with the US and the rest of the world.
- The agreement will have major development scope for our industries, both public and private. US nuclear corporate lobby was decisive force behind this agreement.
- The Agreement placed India's civil nuclear installations under the watch of International Atomic Energy Agency, subject to India's ratification of 'Additional Protocol' which it did recently.



INDIAN NUCLEAR DIPLOMACY

AGREEMENTS WITH OTHER COUNTRIES

- **India signed agreements with France** (France was the first country to sign a civilian nuclear agreement with India **in 2008**, after complete waiver was provided by the NSG), Russia and South Korea to build nuclear reactors.
- **Russia also helped India** in its efforts **to design a nuclear plant** for its submarines. With this India is able to safeguard its long coastline.



INDIAN NUCLEAR DIPLOMACY

AGREEMENTS WITH OTHER COUNTRIES

- India signed agreements with **Mongolia, Canada** (Canada is one of the world's largest exporters of uranium and heavy water nuclear technology), Australia (3rd largest producer of uranium), Namibia and Kazakhstan for the supply of uranium fuel to India.
- Countries like the US, Germany, South Korea, Russia, Australia and UK are looking at **India** as the **largest market** to offer their services, **technology and fuel**.



INDIAN NUCLEAR DIPLOMACY

SIGNIFICANCE OF INDIA'S NUCLEAR AGREEMENTS

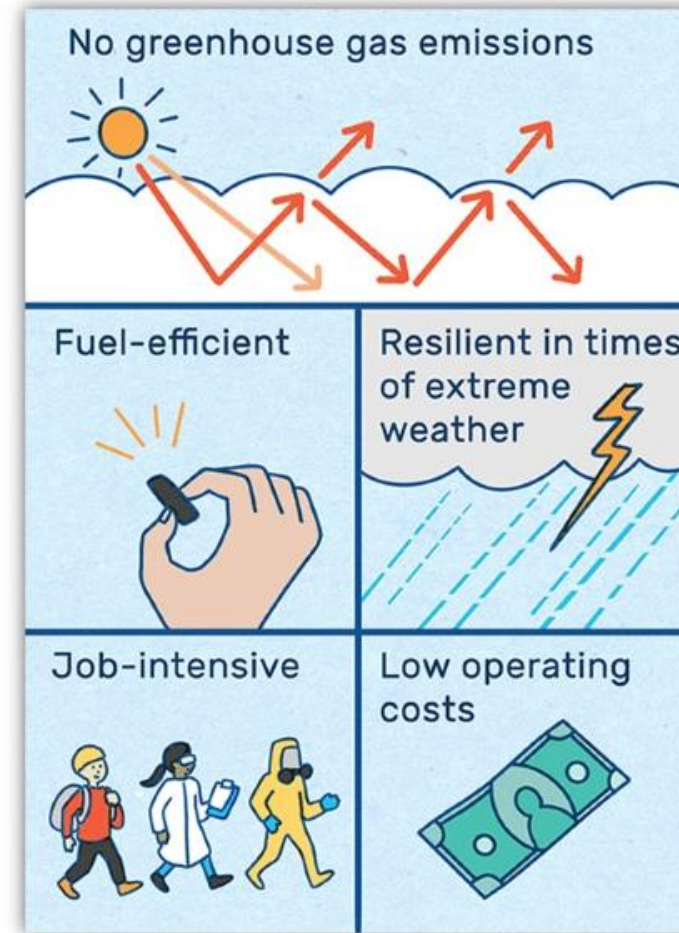
- India is an energy scarce country and most of its energy is derived from the conventional fuels like coal and crude oil. India is heavily dependent on imported crude oil; India imports nearly 72% of its demand which is not only a drain on the foreign reserves but also poses significant environmental risks.
- Also, nearly 70% of the power generated in India is thermal generated. The average efficiency of the thermal power plants in India is one of the lowest in the world also the average CO₂ emissions from these plants is very high.
- Also, the reserves of conventional fuel are limited and not sustainable in the long run, so India is looking for new fuels for power generation and nuclear energy is one of the best options we have.



INDIAN NUCLEAR DIPLOMACY

SIGNIFICANCE OF INDIA'S NUCLEAR AGREEMENTS

- Nuclear fuel is a high energy density fuel, it is comparatively cleaner than the other fossil fuels. The cost of electricity generation from nuclear power is also competitive. These factors make nuclear energy a significant contributor in India's energy needs.
- India also has the largest reserves of Thorium in the world. So, India can make use of its own resources.
- Currently India has a flourishing nuclear programme and expects to generate **63 GW of nuclear power by 2032**. It also aims to supply 25% of its electricity demands from nuclear power by 2050.



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